Table S-physchem. Physicochemical descriptors and drug-likeness rules for the 15 core metabolites (descriptive only)

Metabolite	Appears in	PubChem CID	MW (Da)	Crippen logP	TPSA (Å²)	HBD	HBA	Rotatable bonds	Lipinski violations (of 4)	Veber (RotB ⩽ 10, TPSA ⩽ 140)
Acacetin	docking (Table 3); E39 ADMET list	5280442	284.27	2.88	79.9	2	5	2	0	yes
Apigenin	docking (Table 3); Fig 4B focus; Fig 4A top 10; E39 ADMET list	5280443	270.24	2.58	90.9	3	5	1	0	yes
Dopamine	docking (Table 3); E39 ADMET list	681	153.18	0.6	66.5	3	3	2	0	yes
Genistein	docking (Table 3); Fig 4B focus; E39 ADMET list	5280961	270.24	2.58	90.9	3	5	1	0	yes
Levodopa	docking (Table 3); Fig 4B focus	6047	197.19	0.05	103.8	4	4	3	0	yes
Lithocholic Acid	docking (Table 3); E39 ADMET list	9903	376.58	5.51	57.5	2	2	4	1	yes
Quercetin	docking (Table 3); Fig 4B focus; Fig 4A top 10; E39 ADMET list	5280343	302.24	1.99	131.4	5	7	1	0	yes
5-(3,4-Dihydroxyphenyl)pentanoic acid	Fig 4A top 10	49831816	210.23	1.9	77.8	3	3	5	0	yes
5-(3-Hydroxyphenyl)pentanoic Acid	Fig 4A top 10	10559720	194.23	2.19	57.5	2	2	5	0	yes
Acetate	Fig 4A top 10	175	60.05	0.09	37.3	1	1	0	0	yes
Butyrate	Fig 4A top 10; E39 ADMET list	104775	88.11	0.87	37.3	1	1	2	0	yes
Equol	Fig 4A top 10	91469	242.27	2.82	49.7	2	3	1	0	yes
Kaempferol	Fig 4A top 10; E39 ADMET list	5280863	286.24	2.28	111.1	4	6	1	0	yes
Luteolin	Fig 4B focus; Fig 4A top 10; E39 ADMET list	5280445	286.24	2.28	111.1	4	6	1	0	yes
Myricetin	Fig 4A top 10; E39 ADMET list	5281672	318.24	1.69	151.6	6	8	1	1	no

Computed with RDKit 2026.09.1 on 2026-11-01 from the SMILES in the 228-metabolite list (largest fragment, neutralised; butyrate and acetate are stored as anions). Values are calculated descriptors, not measurements; logP is the Crippen estimate. ADMETlab 3.0 predicted endpoints are given in Table S-ADMET (model outputs, descriptive); SwissADME was not run (its web form cannot be scripted). The table is descriptive and was not used to select metabolites. Gut-derived metabolites are largely endogenous or microbial products and can act locally in the gut lumen, so drug-likeness rules are given for reference only. The 15 compounds are the union of the core metabolites that appear in the main text: the seven compounds of the nine docking pairs (Table 3), the five metabolites of the Fig 4B focus network, the ten highest-degree metabolites of Fig 4A and the ten ADMET-check compounds fixed in E39 (column "Appears in"). Decanoic acid (docking record failed the InChIKey check; pair shown in Table S-docking_excluded_pair) and 7-aminoheptanoic acid (not in the current Fig 4B focus network) are therefore not included. AI preliminary calculation (E65).